



A Foundation-Model T-cell Dysfunction Screen: Translational Candidates, Then a Full Audit

Four checkpoint/persistence hits, tissue-validated -- and what a full concordance, ambient, and donor-level audit does and does not change about them.

Kaisar Dauyey · Shinji Nakaoka

Laboratory of Mathematical Biology, Faculty of Advanced Life Science, Hokkaido University

Contact: k.dauyey.bio.nu@gmail.com

Slides, report and build package: github.com/Kays3/talks

T cells in lung cancer: why they matter, and what was already known

The disease

Lung cancer comes in distinct types. **Small cell lung cancer (SCLC)** is the most aggressive thoracic malignancy; **lung adenocarcinoma (LUAD)** is a non-small-cell type.

SCLC itself splits into subtypes defined by master transcription factors (ASCL1, NEUROD1, POU2F3) (Rudin 2019).

The immune angle

T cells can kill tumour cells. Under chronic stimulation they can become dysfunctional ("exhausted"), marked by inhibitory receptors such as PD-1, TIM-3 and CTLA-4 (Wherry 2015).

Immune checkpoint inhibitors release these brakes, but add only a modest survival benefit in extensive-stage SCLC (Horn 2018), despite its high tumour mutational burden.

The open question

Whether SCLC T cells are dysfunctional, and through which genes, was poorly characterised at single-cell resolution when this work began.

SCLC subtypes differ in immune activation; an inflamed subtype benefits most from immunotherapy (Gay 2021). This talk asks what a model trained on T cells themselves can say.

In plain words: we study the immune cells inside lung tumours, because drugs that re-activate them help SCLC patients less than hoped, and nobody knows exactly why.

Three tools, one analogy each

Single-cell RNA sequencing

Reads which genes are switched on in **each individual cell**, not in a ground-up tissue sample.

Analogy: a list of every fruit in the bowl, instead of the smoothie.

Here: 46,140 T cells from 42 people (HTAN MSK collection, CELLxGENE).

Geneformer, a foundation model

A neural network pre-trained on a large corpus of single-cell profiles (Theodoris 2023). It reads a cell as its genes **ranked by expression**, most-expressed first, and turns it into a list of numbers (an **embedding**) that puts similar cells near each other.

Analogy: a map coordinate for every cell. We fine-tuned it to tell SCLC, LUAD and normal-tissue T cells apart.

In silico perturbation

Delete one gene from a cell's ranked list, or move it to the top ("overexpress"), re-read the cell, and measure how far the model's picture of it moves toward another disease state.

Analogy: a Perturb-seq screen run on the computer. **Nothing is edited in a lab**: it generates hypotheses, it is not a knockout.

In plain words: we taught a model what SCLC, LUAD and normal T cells look like, then asked it "what if this gene were missing, or boosted?" for thousands of real cells.

Our 24JSDP poster (P25) and what the audit changed

Our 24JSDP poster (P25) reported a donor-held-out classifier (91.9% accuracy), four replicated edits (TIM-3, TIGIT, CTLA-4, IL7R) and tissue validation in 5 Visium SCLC sections (15,632 spots).

Poster statement	Status after the audit (this talk)
Four edits concordant, FDR < 0.05 in both arms, same sign in all 3 SCLC donors	Deletion replication stands. The concordance half was scored on different cell sets per arm (overexpress 2,424 cells, delete 202-1,131) and has not been rerun.
"The model implies Normal < SCLC < LUAD on the checkpoint axis ... an open question"	Closed as null: donor-level SCLC vs LUAD, p = 0.216.
Detection vs effect size, rho = -0.60: sparse genes fake big effects	Unchanged; the original talk's own QC slide, shown here as Results R2.
Tissue: dysfunction tracks T-cell abundance (rho 0.161); talk: antigen presentation rho 0.361	Unchanged; not re-analysed here.

In plain words: the poster's main hits survive; one of its claims (an ordering of the three disease states) did not hold up once each person was counted once.

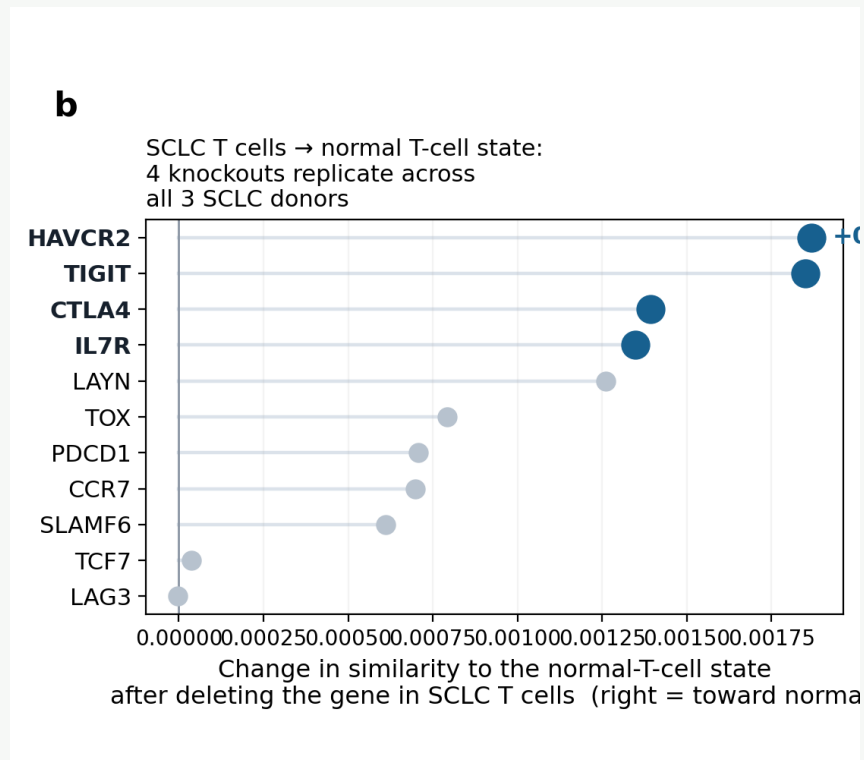
A perturbation screen can look statistically overwhelming and still be wrong

Foundation models fine-tuned on single-cell transcriptomes can be probed for candidate disease-state drivers: delete or overexpress a gene's simulated expression in a held-out cell, and read the shift in a fine-tuned classifier's predicted state. A dose-responsive driver should move the model toward its target when overexpressed and away from it when deleted.

A classifier trained on real tissue also learns whatever else correlates with its labels -- ambient RNA from other cell types, detection-rate differences, cohort composition, tissue of origin. None of that is cell-intrinsic biology, and a screen does not tell you on its own which one it found.

This talk presents a set of translational candidates identified this way, then applies the audit a screen like this needs: a concordance test, a screen-level ambient-contamination check, and a donor-level reanalysis of a cross-cohort ordering claim.

Four replicated checkpoint/persistence hits, tissue-validated



A donor-disjoint SCLC/LUAD/normal T-cell classifier supported an applied in-silico perturbation screen for candidate CAR-T/ICI-relevant drivers. Four hits replicated across the screen: **TIGIT, TIM-3 (HAVCR2), CTLA-4, IL7R.**

Independent Visium spatial validation (GSE263196, 5 SCLC specimens, 15,632 in-tissue spots) supported a tissue-validated antigen-presentation programme.

Presented at JSDP, 2026-08-17. No caveat attached here -- none of these four hits is implicated in anything that follows.

One test was already flagged as pending; one new screen needed the same scrutiny

The poster's own hedge

“The model implies Normal < SCLC < LUAD on the checkpoint axis. This conflicts with the clinical tumour-level picture of SCLC as cold and ICI-resistant, so it **remains an open question rather than a reconciled conclusion.**”

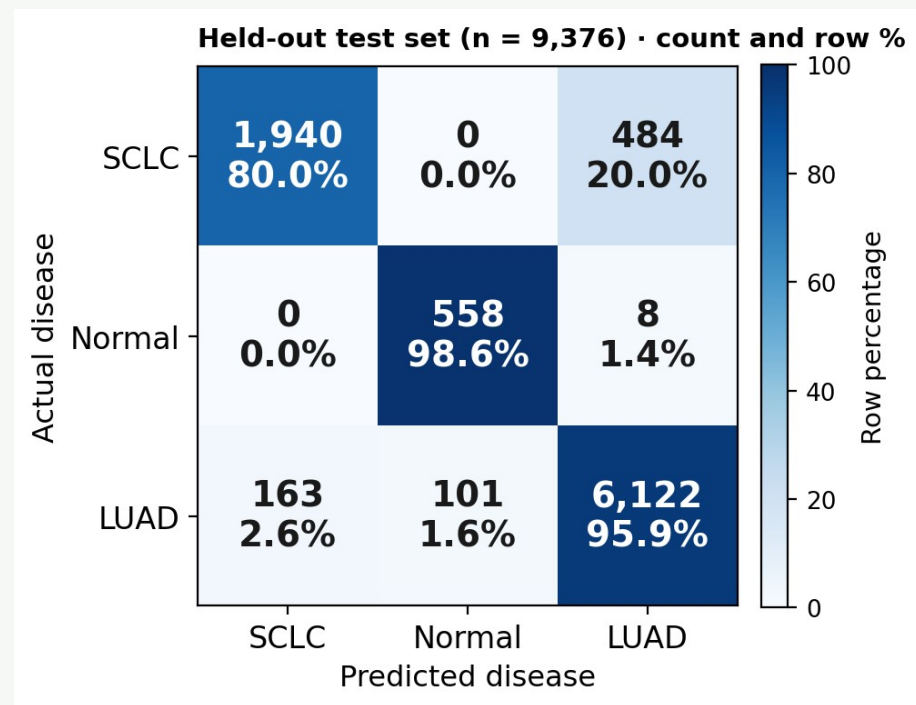
The talk's own QC slide, “TESTED SINCE THE POSTER”

“The states are not on a line, so there is no ordering to contradict... Test 1 of 4 — sign directions, not centroid geometry. **Pending T2 and T3.**” The donor-level test was named as open before this work began.

What this talk adds

A later, expanded screen surfaced two new high-FDR candidates needing the same concordance scrutiny the original four received. This talk **completes the pending test** and applies that scrutiny — it does not reopen a settled claim.

46,140 T cells, 42 donors, one fine-tuned classifier



Held-out test performance

91.9% accuracy, macro F1 0.903, donor-disjoint split. Geneformer V2 104M fine-tune.

Normal-class test performance rests on 1 held-out donor (566 cells)

A third of this macro-F1. Already flagged in the repo's own methods (perturbation_workflow/METHODS.md, lines 57-67 the split table, line 112: "Normal-class metrics are flagged explicitly given 1 test donor").

19 SCLC + 22 LUAD + 4 normal = 45, but 42 unique donors: three donor_ids (RU675, RU682, RU684) each carry two biospecimens from the same clinical patient, one LUAD tumour and one normal-tissue sample (METHODS.md, lines 33-39).

In plain words: shown T cells from 8 people it had never seen, the model named the right disease state for about 92 of every 100 cells; the normal-tissue part of that score rests on one person.

Delete or overexpress, in held-out cells, across six comparisons

Each gene is deleted (its token removed from a cell's rank-ordered input) or overexpressed (promoted to the front of the rank) in held-out cells, and the shift in the classifier's predicted disease-state probability is read for both operations.

Six source→goal disease-state comparisons: SCLC→LUAD, SCLC→normal, LUAD→SCLC, LUAD→normal, normal→SCLC, normal→LUAD.

A dose-responsive driver should move the model toward its target under overexpression and away from it under deletion — the two operations are read together, never one alone.

A real driver must move opposite ways under deletion and overexpression

The concordant criterion, used throughout this repo's perturbation work:

```
delete_fdr < 0.05 AND overexpress_fdr < 0.05 AND delete_shift × overexpress_shift < 0 AND delete_n ≥ 25
```

Significance in one arm is not enough. A gene must clear both arms and reverse sign between them — the signature a dose-responsive driver actually produces. A gene can be significant in both arms and still fail this test if deletion and overexpression push the classifier the **same** direction — the model reacting to the gene's presence at all, not to a dose-responsive relationship with the labelled state.

In plain words: a gene counts only if removing it and boosting it push the cell in opposite directions, each beyond chance; $FDR < 0.05$ means at most about 5% of such calls are expected to be false.

Two further checks, applied after the screen, not instead of it

Ambient-risk audit

Every top-effect gene is checked against a curated ambient-contamination panel and a curated T-cell-intrinsic anchor panel. The flag threshold is the 25th percentile of the full ambient-anchor population's own risk scores — a within-panel, not an absolute, cutoff.

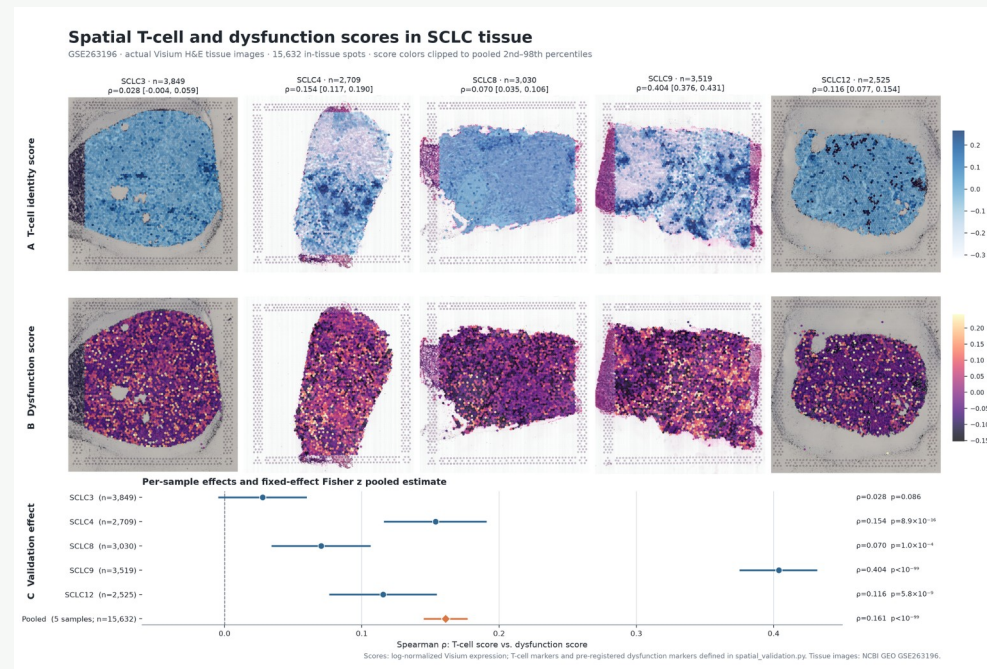
Donor-weighted reanalysis

Any cross-cohort comparison is recomputed donor-level (mean within donor first, then an unweighted mean of donor means) alongside the original cell-weighted statistic, to separate a real group difference from one donor's cell count dominating the mean.

Why both, not either

A significance test alone cannot distinguish a cell-intrinsic effect from an ambient or cell-count artifact. Both audits are applied to every candidate discussed in Results, not selectively to ones that already look questionable.

Four hits replicate and validate independently in tissue



TIGIT, TIM-3 (HAVCR2), CTLA-4, IL7R

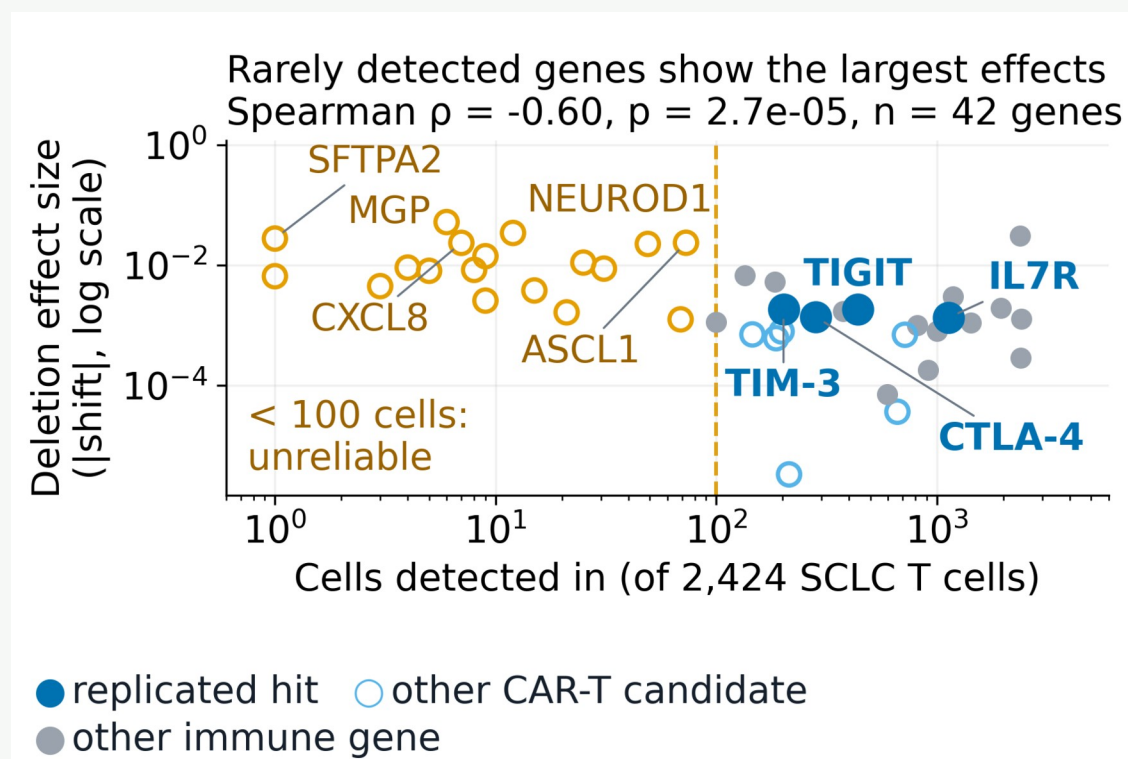
Four replicated checkpoint/persistence edits from the applied screen (screen_b.png, Introduction). Replication here means: deletion FDR < 0.05 and the same deletion sign in all 3 SCLC donors.

GSE263196 Visium, 5 SCLC specimens, 15,632 in-tissue spots: antigen-presentation programme $\rho = 0.361$, 7.95 sigma above the null.

The concordance half of the original call was made by the pre-2026-09-22 panel runner, which overexpressed into all 2,424 SCLC cells while deleting only in the 202-1,131 cells expressing each gene; it has not been rerun with paired cell sets. Unaffected by S100, ambient or T6 below; the spatial number is untouched. A qualifier, not a retraction.

In plain words: in real tumour slices, spots with more antigen-presentation activity also held more T cells ($\rho = 0.361$, where 0 is no relation and 1 is perfect), far beyond chance (7.95 sigma).

Sparse genes fake big effects, and the poster's own ordering fails its own test



Detection vs. effect size: Spearman $\rho = -0.60$ ($p = 2.7e-5$, $n = 42$) across the full 50-gene panel.

“TESTED SINCE THE POSTER” (talk, slide 7): “the states are not on a line, so there is no ordering to contradict... Working reading: a triangle — SCLC $\leq$ Normal < LUAD... Test 1 of 4 — sign directions, not centroid geometry. **Pending T2 and T3.**”

This is the original team's own self-correction, dated before this work began — not a finding of ours.

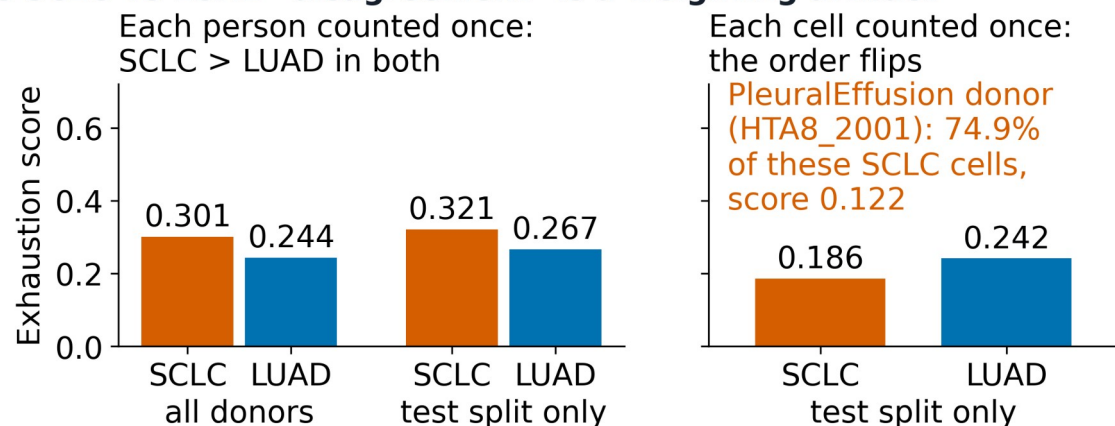
In plain words: the rarer a gene is in the data, the bigger its apparent effect tends to be ($\rho = -0.60$), so the size of an effect alone is not evidence.

Donor-level: no reversal

Complete population: SCLC 0.301 > LUAD 0.244. Test-only: SCLC 0.321 > LUAD 0.267. Two-sided **Monte Carlo** test, 100,000 replicates: $p = 0.216$, 19 vs. 22 donors. **Not significant.**

The open question is now closed as null

The SCLC-vs-LUAD "disagreement" is a weighting artifact



PleuralEffusion: one genuine donor

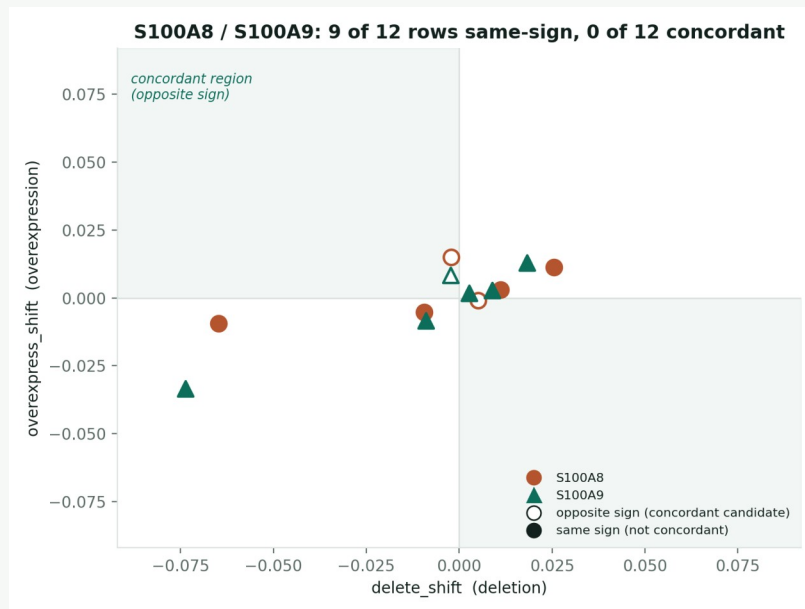
Carries 74.9% of test-only SCLC cells.

HTAN_Participant_ID confirms exactly one participant (HTA8_2001), not several merged (METHODS.md, lines 29-32).

A separate, **exact-enumeration** test on the 3-vs-4-donor test-only slice: only 35 label assignments exist; the smallest attainable two-sided p is 0.0571 (2/35) — that design cannot reach significance regardless of the true effect.

In plain words: counting each person once instead of each cell, SCLC and LUAD T cells do not differ significantly on this axis ($p = 0.216$); one donor had supplied most of the test-only SCLC cells.

S100A8 and S100A9: an overwhelming FDR, and no concordance at all



9 of 12 rows same-sign; 0 of 6 comparisons concordant

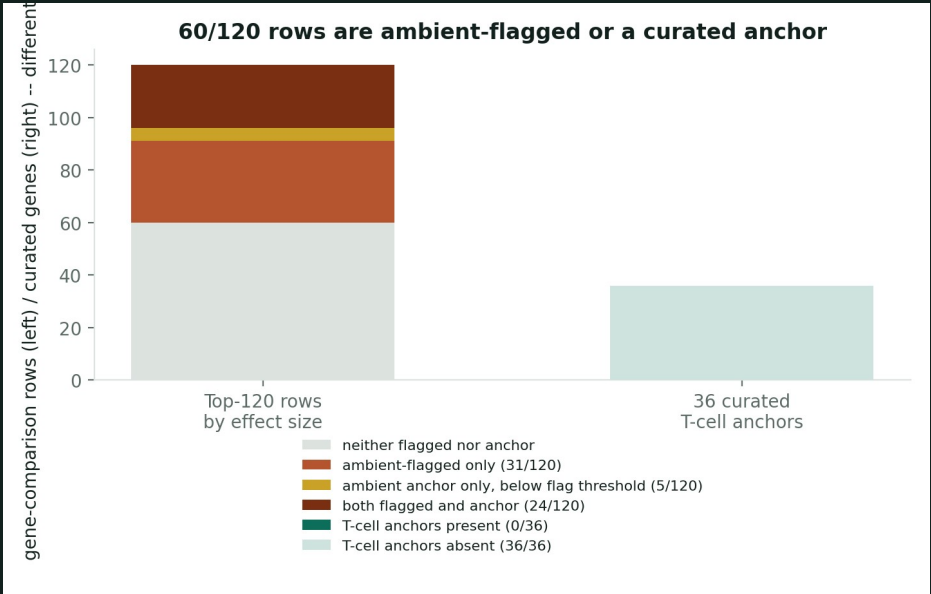
Example: S100A8, LUAD→SCLC deletion — FDR = $1.45e-224$, N = 713 detections, shift = +0.0255. **The FDR is a certainty claim, not a size claim** — it says nothing about direction. First appearance of these two genes in this deck: caveated from the start, not a later correction.

What the LUAD-normal and SCLC-normal rows can and cannot lean on

Of 12 rows, 8 touch the normal class (4 comparisons). For $\text{luad} \leftrightarrow \text{normal}$, 3 of 4 normal donors are the *same patients* as 3 LUAD donors — real protection against a separate-cohort confound, though only 3 of 4. **No such pairing exists for $\text{sclc} \leftrightarrow \text{normal}$.**

In plain words: an FDR of $1.45e-224$ says the shift is almost certainly not chance, not that it is large or meaningful; S100A8/S100A9 still fail the both-directions test in all 6 comparisons.

Half the top 120 is ambient-flagged or a curated ambient anchor



60/120

rows are ambient-flagged, a curated ambient anchor, or both: 31 flagged-only, 5 anchor-only, 24 both — a real partition of the 120, not two overlapping counts added together

5/120

rows are anchors below the flag threshold — mechanical, not a miss: the threshold is the anchors' own 25th percentile of risk, so some anchors falling under it is built in.

0/36

of the 36 curated T-cell-intrinsic anchor genes (a different denominator: the full curated list, not a row count) appear anywhere in the 120.

In plain words: half of the 120 strongest hits carry a flag for possible contamination by RNA from other cells, or are known contaminant marker genes, so they need caution before being called T-cell biology.

One candidate survives every check

NBEAL1

5 of 5 comparisons concordant — every one it appears in — detected in 75.0% of cells.

Passes the same concordance test that eliminates S100A8/S100A9, at a detection rate well above the ambient range flagged for those two. Does not make it a validated driver on its own — makes it the one candidate this screen has not been able to eliminate.

Not everything restored from the original screen fails, and the one thing that doesn't fail is more informative for having survived a real test.

In plain words: NBEAL1 passed the both-directions test in all 5 comparisons it appears in and is seen in 3 of every 4 cells, so it is not a rare-gene artefact; it is a candidate, not a proven driver.

S100 genes did not beat their matched control genes

Question and design

Do simulated deletion and overexpression of S100 genes push LUAD T cells toward normal or SCLC **more than matched control genes** do, and more so for the 4 S100 genes at high ambient-RNA risk (S100A2, S100B, S100A13, S100PBP) than for 4 at low risk (S100A4, S100A6, S100A10, S100A11)?

Each gene vs its own 20 detection-matched controls; the two groups of 4 compared with an exact Mann-Whitney test; all registered in advance.

Result: negative in both comparisons

LUAD → normal: $U = 1.5$, $p = 6/70$ (0.086).

LUAD → SCLC: $U = 5.5$, $p = 34/70$ (0.49).

Stability gate not positive (0/160 leave-one-control-out). No-op check passed. S100P and S100A16 not estimable (40 and 48 cells, below the 50-cell gate). S100A8/S100A9 excluded by design (classifier anchors).

How to read it

The test was small: with 4 v 4 the smallest possible p is $2/70$ (0.029), reached only by complete separation.

The ambient-high genes did not beat matched controls more than the ambient-low genes. No claim is made from the ambient-low genes' high scores.

This **neither confirms nor contradicts** the S100A8/S100A9 slides.

In plain words: in a small, pre-registered test, S100 genes flagged for possible contamination did not stand out against look-alike control genes.

What this establishes, and what it rules out

Establishes

Four checkpoint/persistence hits (TIGIT, TIM-3, CTLA-4, IL7R) replicate and validate independently in spatial tissue data. A concordance test and a screen-level ambient audit are necessary companions to significance for any further perturbation-based driver claim. One further candidate (NBEAL1) clears every check applied.

Resolves

The checkpoint-axis ordering the original team flagged as open (“pending T2 and T3”) is closed: donor-level SCLC-vs-LUAD shows no significant difference in either direction, once a single donor’s cell-count share is accounted for.

Rules out

S100A8/S100A9 as T-cell-intrinsic dysfunction drivers, at high confidence, independent of model precision. Model precision (bf16) as an explanation for any finding in this deck.

What this study does not establish

Detection-confound axis is an imperfect proxy

Ambient risk is scored partly against the classifier being audited — evidence of resemblance to known contaminants, not a direct contamination measurement.

Single cohort; curated lists

One HTAN/CELLxGENE cohort; the ambient-anchor and T-cell-anchor gene lists are curated, not ground truth. A different cohort or anchor set could shift specific gene calls.

Thin normal class, both counts

1 held-out test donor AND 4 cohort donors overall — already a named limitation in the repo's own methods (METHODS.md, lines 24-25), carried forward here, not newly flagged.

Tissue-of-origin skew, per contrast

SCLC: 7/19 (37%) lung, 12/19 metastatic.
LUAD: 16/22 (73%) lung. Normal: 4/4 (100%) lung. Skews
sclc_to_normal/normal_to_sclc, not
luad_to_normal/normal_to_luad (which
benefits from the 3-of-4 shared-patient
pairing above).

Open questions: the original talk's CRISPR TIGIT/HAVCR2 knockout proposal; does NBEAL1 replicate independently; the talk's own still-open question — “if SCLC T cells are not exhausted, what are they, and why does ICI still fail?” A clean negative, honestly obtained, is a contribution.



Terms used in this talk

Term	Meaning here
Single-cell RNA-seq	Measures which genes are active in each individual cell.
T-cell dysfunction / exhaustion	A worn-out T-cell state after chronic stimulation, with inhibitory receptors (PD-1, TIM-3, CTLA-4) (Wherry 2015).
Immune checkpoint	An inhibitory receptor that brakes T cells; checkpoint-inhibitor drugs block it.
Foundation model (Geneformer)	A model pre-trained on many cells, then fine-tuned for one task (Theodoris 2023).
Embedding	The list of numbers the model uses to place a cell; similar cells sit close together.
In silico perturbation	Deleting or boosting a gene in the model's input and measuring the shift.
Concordant	Deletion and overexpression move the cell in opposite directions, both significant.
FDR	False discovery rate: the expected share of false calls among all calls (Benjamini 1995).
Donor-level analysis	Each person counted once, so one person's many cells cannot dominate.
Ambient RNA	Free-floating RNA from other cells that contaminates a cell's reads (Young 2020).
Matched controls	Non-S100 genes chosen to be detected about as often as the gene tested.
Mann-Whitney U	A rank test of whether one group tends to score higher than another (Mann 1947).
Pre-registration	Fixing the test and its pass/fail rule in writing before seeing the result.
bf16	A lower-precision number format; used to check that rounding does not drive results.

Backup slides

Deep-dive technical slides, kept for questions:

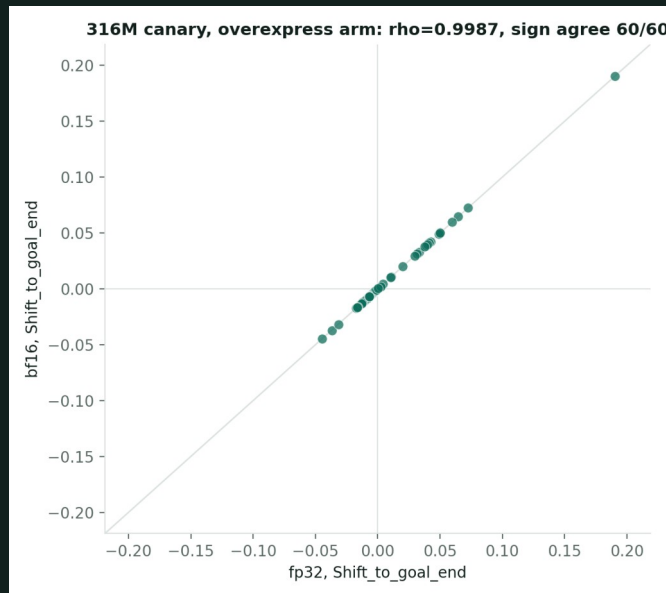
A1 · Precision control: is any of this a bf16 artefact? (was slide 15 of the 25 Sep deck)

A2 · Mechanism: the four hits in one interaction neighbourhood (was slide 16; the designated cut for a shorter slot)

A3 · S100 ISP per-gene scores (new, 2026-09-28)

A4 · All 31 immune / lineage genes ranked by deletion shift (panel b of the detection figure, moved here so it can be read at 14 pt)

No: precision replicates cleanly; model size is the axis that moves things



Same model, fp32 vs. bf16 (half precision)

104M panel (300 rows): rho = 0.9998, sign agreement 298/300. 316M canary (60 rows): rho = 0.9987, sign agreement 60/60.

The pre-registered 104M sign-agreement gate is still recorded **FAIL** — a bitwise-deterministic fp32-vs-fp32 rerun made the noise floor zero, not a usable sign threshold; the domain reviewer's ruling was that the frozen gate stands unamended, not that bf16 disagrees with fp32. Both are true at once.

Same model, changed precision vs. changed size

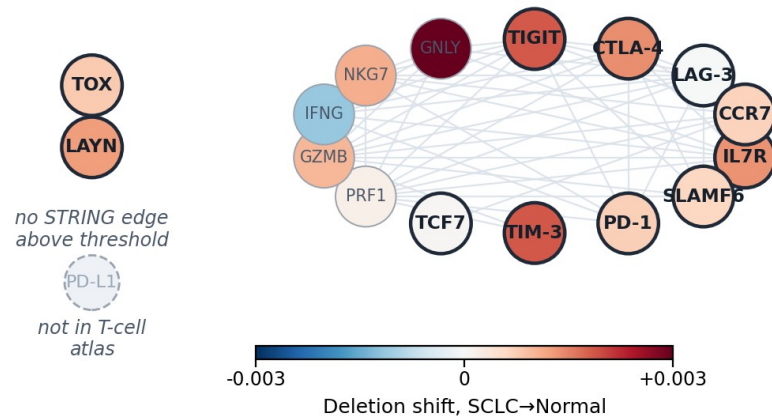
Precision (104M fp32→bf16): rho = 0.9998. Size (104M→316M, both bf16): rho = 0.489. Using (1-rho) as the divergence measure, size moves results ~2,500x (~3.4 orders of magnitude) more than precision. None of this deck's findings are precision artifacts.

In plain words: rerunning the model at lower numerical precision gave essentially the same answers (rho 0.9998), so rounding does not explain any result in this talk.

The four original hits sit in one connected interaction neighbourhood

C

All 11 candidates on the STRING map;
bold = panel b candidate,
colour = its SCLC→Normal shift



From prior literature, not inferred from these cells

TIGIT: CAR-T support. TIM-3: solid-tumour knockout precedent. CTLA-4: real signal, less CAR-T-specific. IL7R: persistence, not exhaustion. 54 STRING edges among 16 context genes; TOX and LAYN have no edge above threshold; PD-L1 is absent from the T-cell atlas.

No caveat — these four hits are unaffected by anything in this deck's audit.

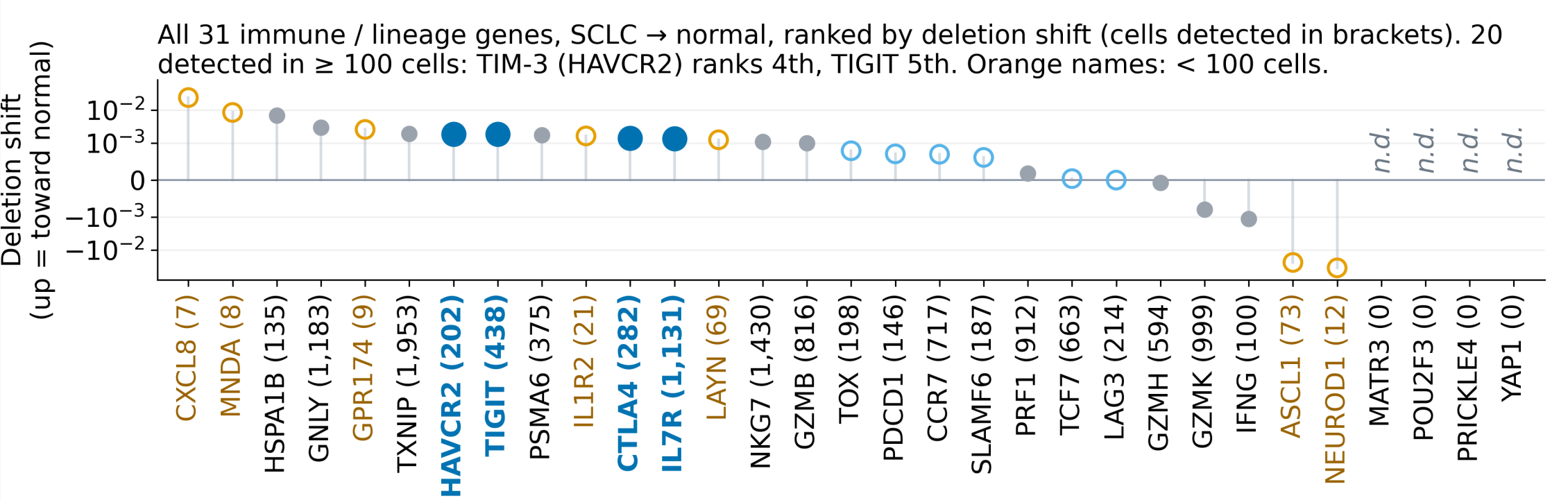
In plain words: the links between these genes come from STRING, a public database of known and predicted protein interactions (Szklarczyk 2023), i.e. from prior literature, not from these cells.

Each S100 gene's percentile against its own 20 matched controls (Q)

Gene	Ensembl ID	Group	Ambient risk	Q, LUAD→normal	Q, LUAD→SCLC
S100A2	ENSG00000196754	ambient-high	0.955	88.1	97.6
S100B	ENSG00000160307	ambient-high	0.859	21.4	16.7
S100A13	ENSG00000189171	ambient-high	0.796	26.2	59.5
S100PBP	ENSG00000116497	ambient-high	0.744	54.8	78.6
S100A4	ENSG00000196154	ambient-low	1.1e-06	88.1	97.6
S100A6	ENSG00000197956	ambient-low	3.4e-05	97.6	88.1
S100A10	ENSG00000197747	ambient-low	0.000312	92.9	88.1
S100A11	ENSG00000163191	ambient-low	0.000144	64.3	50.0

Group test (exact two-sided Mann-Whitney, 4 v 4): LUAD→normal U = 1.5, p = 6/70; LUAD→SCLC U = 5.5, p = 34/70. Negative in both. The ordering seen (ambient-low above) is descriptive only; the ambient-low genes' high Q was not tested and supports no claim.

Every immune / lineage gene, ranked by its deletion effect



In plain words: the four replicated hits sit near the top among well-detected genes; the very top is taken by genes seen in fewer than 100 cells, whose effects are unreliable.